
Research Proposal

MSc Thesis in Data Science and Business Analytics

Dutch Healthcare Accessibility: A Three Tiered Gatekeeping System Enabling Acute Healthcare Access; a Geographic Analysis

by

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Introduction Geospatial analysis has proven to be a crucial tool in assessing the global and local accessibility of healthcare systems worldwide. Through analyzing transportation networks, population density and location, and availability of local healthcare providers studies have been able to identify regions and population points with poor and very poor healthcare accessibility and provide recommendations for improvement (Lawl and Anyiam, 2019).

Furthermore, the geospatial accessibility of population points to GPs in the Netherlands has been investigated before, showing that the Netherlands has substantial accessibility to its first tier of healthcare by bike and car (van de Wal, 2024). While providing a methodology for further investigation in the Netherlands, this paper did not show the accessibility of Dutch population points to acute healthcare, nor the interaction between the three-tiers of the healthcare system, thus leaving a significant portion of the story open for further study.

The Dutch healthcare system is designed to ensure accessibility of emergency care to its residents by gatekeeping through different filters (Ministry of Health, Welfare and Sport, 2016). To ensure that only necessary cases are seen at Spoedpost Hulp clinics, Dutch residents are first required to be recommended to a specialist by their GP. During the nights and weekends residents may contact their local Huisartsenpost for such a recommendation. In the case of a life-threatening emergency residents instead contact the emergency number and an ambulance may instead take them to a Spoedpost Hulp immediately. This ensures availability of specialists at the final tier, while still allowing for a direct route when necessary.

This contrasts with some similar EU countries such as Portugal where the emergency tier may be the first stop for most people who are in need of any sort of medical help (Joosten and Gromicho, 2026). Portugal has the additional factor that healthcare is completely free and open for all people. This may introduce a confounding effect as there is no financial incentive for Portuguese residents to not go immediately to the emergency room. Whereas in the Netherlands, the standard GP is covered by basic insurance while residents may have to pay out of their own risk should they need to see a specialist such as at an emergency clinic.

Overall, this thesis seeks to tell the story of acute healthcare accessibility in the Netherlands considering the process through which acute healthcare may be accessed in a geospatially heterogeneous setting.

Research questions How can we explain to whom, where, when, and why interactions between the three tiers of the Dutch Healthcare System enable or prevent accessibility to acute healthcare services using geospatial analytics on open-source data?

Methodology and Techniques Gather and merge datasets from across open sources. This should include CBS Netherlands geographic-population data, Open Street Map geographic-transportation network data, and Zorgkaart Nederland geographic-healthcare facility data. Use gravity style metrics to model healthcare accessibility as in van de Wal, 2024. Visualize areas of good and poor accessibility on city, municipality, and country level granularities. Analyze and visualize the sensitivity of accessibility to shocks to healthcare facility availability.

Data and/or expected results A thorough analysis of transportation networks, population points (including location and size), and location and size of healthcare facilities will be conducted. This data will be gathered from open sources including Open Street Maps (OpenStreetMap contributors, 2026), CBS (Centraal Bureau voor de Statistiek, 2024), and Zorgkaart Nederland respectively (Zorgkaart Nederland, 2026).

Higher resolution population grid data may also be investigated from Meta's AI for Good: High Resolution Population Density Maps and Demographic Estimates (Meta, n.d.) or the similar population density data from WorldPop (Bondarenko, et al., 2025).

The Netherlands' GPs have been shown to be highly accessible in previous geospatial studies (van de Wal, 2024). However, this does not mean that it is in fact totally accessible. I expect that locations with low GP accessibility may also have low accessibility of acute healthcare, however whether there are locations with high GP accessibility but low acute accessibility will also be investigated. Due to the heterogeneity of population densities, development, and healthcare outcomes geospatial heterogeneity among acute healthcare accessibility may also be expected.

This thesis seeks to take an analytical, atomistic approach to descriptive geospatial analytics. The main deliverables for this thesis should include relevant visualizations of where in the Netherlands accessibility to acute healthcare changes and where in the Netherlands the impact of the underlying system changes, as investigated by the sensitivity of healthcare accessibility simulation to changes in availability of facilities.

Literature Joosten, A. and Gromicho, J., 2026. *Health system design, incentives, emergency care, and structural workload: Comparative research notes on Portugal and the Netherlands*. Unpublished manuscript. Centro Hospitalar Barreiro-Montijo and Amsterdam Business School, University of Amsterdam.

Krishnakumari, P.K., den Hertog, D. and Gromicho, J., 2026. *Hands-on geospatial analytics for the social good. Volume one: Data journeys*. Unpublished manuscript.

van de Wal, M., 2024. *Socio-spatial accessibility of general practitioners in the Netherlands*. Master's thesis. Utrecht University.

Ministry of Health, Welfare and Sport (2016) *Healthcare in the Netherlands*. The Hague: Ministry of Health, Welfare and Sport.

Lawal, O. and Anyiam, F. E. (2019) 'Modelling geographic accessibility to Primary Health Care Facilities: combining open data and geospatial analysis', *Geo-spatial Information Science*, 22(3), pp. 174–184. doi: 10.1080/10095020.2019.1645508.

OpenStreetMap contributors, 2026. *Planet dump retrieved from <https://planet.osm.org>*. OpenStreetMap Foundation. Available at: <https://www.openstreetmap.org> (Accessed: 5 March 2026).

Centraal Bureau voor de Statistiek, 2024. *Population statistics 100 m grid Netherlands*. Available via: PDOK. Available at: <https://www.pdok.nl> (Accessed: 5 March 2026).

Zorgkaart Nederland, 2026. *Healthcare facility listings for the Netherlands*. Patiëntenfederatie Nederland. Available at: <https://www.zorgkaartnederland.nl> (Accessed: 5 March 2026).

Bondarenko, M., Priyatikanto, R., Tejedor-Garavito, N., Zhang, W., McKeen, T., Cunningham, A., Woods, T., Hilton, J., Cihan, D., Nosatiuk, B., Brinkhoff, T., Tatem, A. and Sorichetta, A. (2025) *Constrained estimates of 2015–2030 total number of people per grid square at a resolution of 3 arc seconds (approximately 100 m at the equator), R2025A version v1*. Global Demographic Data Project. Southampton: WorldPop, School of Geography and Environmental Science, University of Southampton. Available at: <https://doi.org/10.5258/SOTON/WP00839>.

Meta (n.d.) *High resolution population density maps*. AI for Good datasets. Available at: <https://ai.meta.com/ai-for-good/datasets/high-resolution-population-density-maps/> (Accessed: 9 March 2026).

Timetable

1st Month

Activity	Date
Discuss Research Proposal/Methodology	1. March 9, 2026
Discuss Literature Review	2. March 23, 2026

2nd Month

Activity	Date
Discuss Initial Descriptive Analysis	3. April 6, 2026
Discuss Sensitivity Analysis	4. April 20, 2026

3rd Month

Activity	Date
Discuss results and conclusions of study	5. May 4, 2026
Discuss Thesis Draft and Revision	6. May 18, 2026

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Master's Thesis Data Science
Master's Thesis Business Analytics

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Supervisor
Joaquim Gromicho

Signature

Second marker (suggestion by supervisor)
Jeroen de Mast

Thesis coordinator
C.G.H. Diks

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